

Public Economics Practicum 2

Discussion Questions:

	8th edition	9th edition	10th edition
Chapter 5	2, 3a, 3d, 8, 11	4, 6a, 6d, 11, 13	4, 7a, 7d, 12, 14

Additional Exercises:

- 2.1 Anne and Beatrice are neighbours. Beatrice loves pets, and considers adopting either a cat or a dog. Anne is very scared of dogs, and would prefer to stay inside her house permanently if her neighbour would adopt a dog. Consider the following payoffs for Anne and Beatrice.

		Beatrice	
		cat	dog
Anne	stay inside	150, 200	150, 350
	go outside	450, 200	0, 350

- (a) If Anne and Beatrice decide independently, what is the Nash equilibrium? Is this an efficient outcome?
- (b) Suppose Anne and Beatrice can make binding agreements, what will they do?

Now suppose that Anne's payoff of staying inside after Beatrice adopts a dog is 350.

- (c) If Anne and Beatrice decide independently, what is the Nash equilibrium? Suppose Anne and Beatrice can make binding agreements, what will they do? Explain why this differs from the difference between (a) and (b).

- 2.2 A firm produces Q liters of paint at costs $10 + Q^2$. The firm sells its paint at a fixed price of 10 per liter. Emissions, like Volatile Organic Components, are given by the function $W = \frac{1}{4}Q$. Environmental damage D due to these emissions is estimated at $D = 4W^2$.

- (a) Show in one figure the marginal private benefit, the marginal private cost, and marginal external damage curves.

- (b) Compare the privately optimal outcome and the socially efficient outcome in terms of output, profits, and damage.
- (c) What is the level of the Pigouvian tax?
- (d) Now suppose that the firm reduces its costs to $10 + cQ^2$, where $c < 1$. Show that the Pigouvian tax increases when c decreases. Explain why.

2.3 A firm located on a riverbank produces Q tons of chemicals, which it can sell at a price of P per ton. Production costs equal $\frac{1}{2}cQ^2$. Production pollutes the river. The level of pollution in the river is given by $W = \frac{1}{2}Q$. Downstream, a farmer uses the water from the river to irrigate his land. To prevent damage to his crops, he must have clean water. Removing pollution W from the river costs $D = 2dW^2$. Assume that $P > 0$, $c > 0$, and $d \geq 0$.

- (a) Show in one figure the marginal private benefit, the marginal private cost, and marginal external cost curves.
- (b) Suppose that the firm has the property rights to the river and that bargaining is costless. What is the outcome of bargaining between the firm and the farmer, in terms of production and payments from one party to the other?
- (c) How much does social efficiency increase due to the agreement you found under b), compared to the case without agreement? How does this increase in social efficiency depend on d ?
- (d) Is the outcome of the bargaining different if the farmer has the property rights to the river?

2.4 Suppose that there are two firms, firm 1 and firm 2. Currently, both firms emit 1000 ton CO₂. The firms differ in their cost of *reducing* emissions, $C(E)$, where E is the reduction in tons CO₂. These costs, which include both lower production and measures to make production less polluting, are given by, respectively

$$C_1(E_1) = \frac{1}{6} (E_1)^2$$

$$C_2(E_2) = \frac{1}{4} (E_2)^2$$

The government has decided that the CO₂ emissions of these firms together should be reduced to 1500 ton. Hence $E_1 + E_2 = 500$.

- (a) Suppose the government uses a Cap-and Trade policy (with the cap at 1500 ton). What is the price that arises on the ‘market’ for a permit to emit a ton CO₂? (hint: a picture a la Figure 5.10 may help)

- (b) What is the difference in total cost of reducing emissions between a policy that demands that both firms reduce emissions by 250 ton and the Cap-and-Trade policy?
- (c) Does it matter how the permits are initially distributed among the two firms?

Now suppose that the cost of CO2 emissions to society increase. The government considers to demand even lower total emissions of the two companies. Denote the total reduction by R : $R = E_1 + E_2$. Assume that the marginal external cost of emissions are equal to d euro per ton CO2, where $d > 100$.

- (d) Suppose the government uses a Cap-and Trade policy. What is the socially efficient Cap on total emissions? How does the price that arises on the ‘market’ for a permit to emit a ton CO2 depend on d ?
- (e) Now suppose the government demand that both firms reduce their emissions by the same amount ($E_1 = E_2 = \frac{1}{2}R$). What is the socially efficient level of emissions? Explain the difference with question d).

2.5 University education has positive externalities. Suppose that total demand for university education in a country is given by $P = 100 - 2q$, where P is the price of university education and q the total quantity of university education. Furthermore, *marginal* external benefits of university education are given by $MEB = 25 - \frac{1}{2}q$. Marginal cost of university education is constant and equal to 20, which is also the price of university education when the government does not intervene in this market.

- (a) Derive the socially efficient level of university education, and the level of university education that arises without government intervention.
- (b) What are the total expenditures on a Pigouvian subsidy on university education?

Now suppose that the government decides to provide university education for free (= provides a subsidy of 20 per unit).

- (c) Calculate the efficiency gain or loss that arises if university education is provided for free, compared to the case where there is no government intervention at all. Explain why efficiency increases or decreases. (Hint: draw a figure)

Now suppose that the private returns to education increase, leading demand to change to $P = a - 2q$, where $a > 100$.

(d) Derive how the Pigouvian subsidy depends on a . Explain.

2.6 Consider a town with 5000 identical citizens. Each citizen owns a bike and a car. Each citizen must decide whether to travel to work by bike or by car. Travelling by bike would take $m_f = 40$ minutes. Travel time by car depends on whether there are traffic jams. A traffic jam arises when more than 500 citizens take the car. Hence, travelling by car would take $m_c = 25$ minutes if the number of cars $C \leq 500$. Travelling by car would take $m_c = 25 + \frac{1}{100}(C - 500)$ minutes if $C > 500$. Assume that each citizen wants to minimise own travel time and that every citizen decides independently whether to travel by bike or by car.

- (a) Determine the number of citizens that travel by car in the Nash equilibrium.
- (b) Prove that in the Nash equilibrium, inefficiently many citizens travel by car. Explain why this happens.

A politician argues that traffic jams can be reduced by investing in road capacity. The politician proposes to increase the number of lanes on the road, such that traffic jams only arise if there are more than 2000 cars on the road. After this investment, travel time by car would be $m_c = 25$ minutes if the number of cars $C \leq 2000$ and $m_c = 25 + \frac{1}{100}(C - 2000)$ if $C > 2000$. Another politician argues that it would be better to invest in better bike lanes instead. At similar investment cost, it would be possible to reduce travel time by bike to $m_f = 35$ minutes.

- 1. (c) Calculate the reduction in total travel time for each of these proposals. Compare with the outcomes under a) and explain.

Other types of government interventions are possible. Consider the choice between a Pigouvian subsidy for citizens who travel by bike and a Pigouvian tax for citizens who travel by car.

- 1. (d) Suppose we only care about efficiency in citizens' choice between travel by bike or by car. Discuss which of these two interventions is better and why.

2.7 (this is an old exam exercise with propositions, which can be true or false. It will not be discussed in the tutorial, but serves as extra practice material)

A village has 10 identical rich inhabitants, who all have 1000 euro they do not need to spend directly. Every inhabitant has two options: saving the money or buying a sheep for 1000 euro. Saving yields an interest rate of 10 percent per year. A sheep can be placed on the field next to the village, and can be sold after a year to a butcher. Taking care of a sheep is costless. The price P that an inhabitant gets for his sheep after a year depends on the total number of sheep S on the field:

$$P = 1700 - 100S$$

The field is part of the village, but has no owner, and can only be used for holding sheep. Inhabitants of the village decide independently between saving and buying a sheep

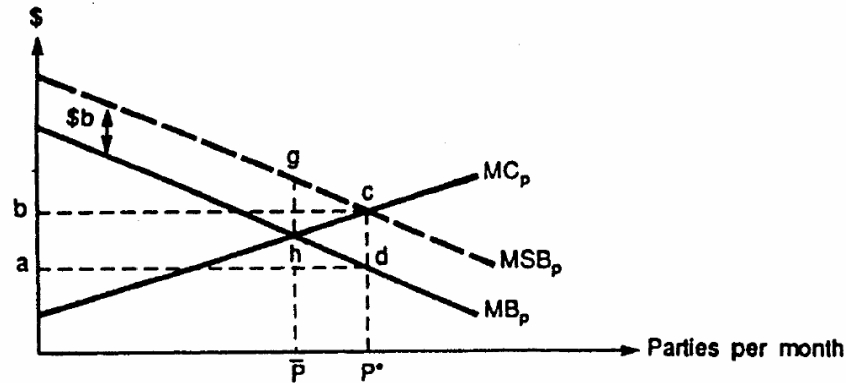
- (a) The Nash-equilibrium number of sheep is 7.
- (b) The socially efficient number of sheep is 3.
- (c) The ‘tragedy of the commons’ arises when resources are non-rival and excludable.

Suppose there is an eleventh inhabitant of the village. This person is poor and does not have 1000 euro to save. The marginal utility of income of the poor inhabitant is higher than the marginal utility of income of the 10 rich inhabitants. The social welfare function of the village is equal to the sum of utilities of all inhabitants (additive). Suppose that the poor inhabitant finds a document that proves that he actually owns the field in the village. This means that he now can ask for a payment from rich inhabitants who want to put a sheep on the field.

- (d) The poor inhabitant can earn maximally 900 euro per year with exploiting the field.
- (e) Social welfare is higher after the poor inhabitant becomes owner of the field.
- (f) If the social welfare function is additive, then a Pareto-improvement always leads to higher social welfare.

Answers to exercises week 2

Rosen 5.2 / 5.4



- The number of parties per month that would be provided privately is $\bar{P}$.
- See schedule MSB_p .
- P^* . Give a per-unit subsidy of \$b per party.
- The total subsidy = abcd. "Society" comes out ahead by gchd, assuming the subsidy can be raised without any efficiency costs (no 'excess burden' or 'deadweight loss'). (Cassanova's friends gain gchd; Cassanova loses chd but gains abcd, which is a subsidy cost to government.)

Rosen 5.3 / 5.6 / 5.7

- Given that there are only a few neighbors, the costs of negotiation should be fairly small.
- Too many internet users to negotiate optimal agreement (i.e. transaction cost are too high).

Rosen 5.8 / 5.11 / 5.12

Privately optimal level of X is given by $MPB = MPC$, or $10 - X = 5$, which gives $X = 5$. The socially efficient level is given by $MPB = MPC + MD$, or $10 - X = 5 + 2$, which gives $X = 3$.

The gain to society is the area of the triangle whose base is the distance between the efficient and actual output levels, and whose height is the difference between private and social marginal cost at the efficient level. Hence, the efficiency gain is $\frac{1}{2}(5 - 3)(7 - 5) = 2$. Alternatively: integrate Marginal net social benefits $MPB - MPC - MD = 3 - X$ to get total efficiency $3X - \frac{1}{2}X^2$. Substituting for $X = 5$ and $X = 3$ yields a gain of $(9 - 4.5) - (15 - 12.5) = 2$.

The optimal (or Pigouvian) tax is equal to the Marginal Damage at the socially efficient level. Here, this is 2. Hence, the tax raises $2 \cdot 3 = 6$ in revenue.

Rosen 5.11 / 5.13 / 5.14

If marginal costs turn out to be lower than anticipated, cap-and-trade achieves too little pollution reduction and an emissions fee achieves too much pollution reduction. With an inelastic marginal social benefit function, cap-and-trade is not too bad from an efficiency standpoint, while an emissions fee causes pollution reduction to be much greater than the efficient level when marginal cost is lower than anticipated. When marginal social benefits are elastic, the opposite is true.

Exercise 2.1

- a) Dog is a dominant strategy for Beatrice. Given that Beatrice will choose dog, Anne will choose to stay inside. Hence, the Nash-equilibrium is (Anne stay inside; Beatrice dog). This is not the efficient outcome. Total payoff is 500, whereas (Anne go outside; Beatrice cat) yields a total payoff of 650.
- b) They will commit to (Anne go outside; Beatrice cat). Anne pays Beatrice $150 \leq P \leq 300$.
- c) The Nash-equilibrium is still (Anne stay inside; Beatrice dog). However, this is now Pareto-efficient. Hence, they will not sign any agreement (or: the agreement will yield the same outcome, without any monetary transfers between Anne and Beatrice). In other words, there is no Pareto-improving agreement possible.

Exercise 2.2

- a) Marginal private benefit (here marginal revenue) $MPB = 10$. Marginal private cost $MPC = 2Q$. Total damage is given by $D = 4(\frac{1}{4}Q)^2 = \frac{1}{4}Q^2$. Hence, marginal damage is given by $MD = \frac{dD}{dQ} = \frac{1}{2}Q$. (You can also calculate marginal damage by using the chain rule).
- b) Privately optimal level of Q given by $MPB = MPC$ (marginal revenue equals marginal cost). This gives $Q^p = 5$, which yields $\pi^p = 15$ and $D^p = 6\frac{1}{4}$. (note: this is of course equal to maximising profit, which is given by $\pi = 10Q - [10 + Q^2]$). Socially efficient level of Q given by $MPB = MPC + MD$, which gives $Q^s = 4$, leading to $\pi^s = 14$ and $D^s = 4$.
- c) Pigouvian tax equals MD at social optimum $Q^s = 4$. Hence, optimal tax equals 2. Note: with a tax of 2 per unit, the firm indeed optimally chooses to produce $Q = 4$ (profit $10Q - 10 - Q^2 - 2Q$ is maximal at $Q = 4$).
- d) Now, $MPC = 2cQ$, so that $MPB = MPC + MD$, yields $10 = 2cQ + \frac{1}{2}Q \rightarrow Q^s = \frac{10}{2c + \frac{1}{2}} = \frac{20}{4c + 1}$ (note: if c goes down, Q^s increases). Hence, Pigouvian tax $Pt = \frac{1}{2} \frac{20}{4c + 1} = \frac{10}{4c + 1}$. This expression decreases when c increases: $\frac{\partial Pt}{\partial c} = -\frac{40}{(4c + 1)^2} < 0$. Hence, if c decreases, the Pigouvian tax should go up. The reason is that here, dam-

age increases quadratically in Q , so that MD increases in Q . Hence, if the socially efficient Q increases (here: due to a reduction in c), the Pigouvian tax also increases.

Exercise 2.3

a) Marginal private benefit (here: marginal revenue) is given by $MPB = P$. Marginal private cost is $MPC = cQ$. Damage is given by $D = 2d(\frac{1}{2}Q)^2 = \frac{1}{2}dQ^2$. Marginal Damage equals $MD = dQ$.

b) Without bargaining, the firm produces $Q = \frac{P}{c}$. At $Q = \frac{P}{c}$, $MPB = MPC$ (Again, this is the same as maximising profit $PQ - \frac{1}{2}cQ^2$ with respect to Q , which yields first-order condition $p - cQ = 0$). Hence, marginal profit is zero at $Q = \frac{P}{c}$, while marginal damage at that level of Q is $d\frac{P}{c} > 0$. Hence, the farmer can compensate the firm for a reduction in Q . This is possible as long as marginal profit (which equals $MPB - MPC = P - cQ$) is smaller than marginal damage. This holds until $P - cQ = dQ$, or $Q = \frac{P}{c+d}$, the socially efficient level. (Note: in the absence of external cost, $d = 0$, the socially optimal level is the same as the privately optimal level).

Comparing $Q = \frac{P}{c}$ with $Q = \frac{P}{c+d}$, we see that the firm's profit ($\pi = PQ - \frac{1}{2}cQ^2$) drops from $\pi^P = \frac{1}{2c}P^2$ to $\pi^S = \frac{P^2}{c+d} - \frac{cP^2}{2(c+d)^2}$. The damage to the farmer decreases from $D^P = \frac{1}{2}d\left(\frac{P}{c}\right)^2$ to $D^S = \frac{1}{2}d\left(\frac{P}{c+d}\right)^2$. Hence, both the farmer and the firm agree on a deal where the firm produces $Q = \frac{P}{c+d}$ and the farmer pays an amount between $\frac{1}{2c}P^2 - \left(\frac{P^2}{c+d} - \frac{cP^2}{2(c+d)^2}\right)$ and $\frac{1}{2}d\left(\frac{P}{c}\right)^2 - \frac{1}{2}d\left(\frac{P}{c+d}\right)^2$ to the firm.

Using a numerical example: If $P = 12$, $c = 1$, and $d = \frac{1}{2}$, we get $\pi^P = 72$, $\pi^S = \frac{144}{2} - \frac{144}{2\left(\frac{3}{2}\right)^2} = 64$, $D^P = 36$, and $D^S = 16$. Hence, the outcome is that the firm produces $Q = 8$ and the farmer pays an amount between 8 and 20 to the firm

c) Without bargaining, social efficiency is $\pi^P - D^P = \frac{1}{2c}P^2 - \frac{1}{2}d\left(\frac{P}{c}\right)^2$. With Coasian bargaining, this is $\pi^S - D^S = \frac{P^2}{c+d} - \frac{cP^2}{2(c+d)^2} - \frac{1}{2}d\left(\frac{P}{c+d}\right)^2 = \frac{P^2}{2(c+d)}$. Hence, bargaining increases social efficiency by $\frac{P^2}{2(c+d)} - \left(\frac{1}{2c}P^2 - \frac{1}{2}d\left(\frac{P}{c}\right)^2\right) = \frac{1}{2}\frac{d^2P^2}{c^2(c+d)}$.

The effect of d on the increase in social efficiency can be found by differentiating this term: $\frac{\partial}{\partial d}\left(\frac{1}{2}\frac{d^2P^2}{c^2(c+d)}\right) = \frac{1}{2}\frac{dP^2}{c^2}\frac{2c+d}{(c+d)^2} > 0$. This is intuitive: an increase in d implies a larger marginal external cost of production, implying an even larger difference between the private and social cost of production.

d) Now, without bargaining the firm is not allowed to produce: $Q = 0$. Then, marginal damage is zero, whereas marginal profits at $Q = 0$ are P . Hence, the firm can (and is willing to!) compensate the farmer for allowing him to produce, as long as marginal profit exceeds marginal damage. This holds until $Q = \frac{P}{c+d}$.

With $Q = 0$, profits and damage are zero; with $Q = \frac{P}{c+d}$, profits equal $\pi^S = \frac{P^2}{c+d} - \frac{cP^2}{2(c+d)^2}$ and damage equals $D^S = \frac{1}{2}d\left(\frac{P}{c+d}\right)^2$. Hence, both parties agree with a deal

where $Q = \frac{p}{c+d}$ and the firm pays the farmer an amount between D^S and π^S . In the numerical example, this is an amount between $D^S = 16$ and $\pi^S = 64$.

Comparing b) and d), we see that in both cases, the socially efficient level arises. However, property rights do affect the distribution of rents.

Exercise 2.4

a) Total reduction in emissions, $E_1 + E_2$, must be equal to 500. Marginal cost of reducing CO2 for firm 1 equals $MC_1 = \frac{1}{3}E_1$, and for firm 2 $MC_2 = \frac{1}{2}E_2$. In equilibrium, it must be that $MC_1 = MC_2$ and the price of a permit to emit 1 ton CO2 P must be equal to marginal cost (see Rosen page 89-90). Hence, we have $\frac{1}{3}E_1 = \frac{1}{2}E_2$ and $E_1 + E_2 = 500$. Solving gives $E_1 = 300$ and $E_2 = 200$. Hence, the price for a permit equals 100.

b) Total cost of the Cap-and-Trade system is 25000. Total cost of demanding both firms to reduce emissions by 250 equals $26041\frac{2}{3}$.

c) It does not matter how the permits are distributed for the reduction in emissions of each firm (unless the government sells permits at $P > 100$, in which case total emissions < 1500). However, the initial distribution of permits does affect transfers. Clearly, permits will not be bought and sold if firm 1 gets 700 permits and firm 2 gets 800. However, if, for instance, both firms get 750 permits, then firm 1 will sell 50 permits to firm 2, and receive (maximally) 5000 in return.

Note that if the government sells the permits rather than giving them for free, the government obtains some revenue (at the expense of the firms).

d) Let $R = E_1 + E_2$. Marginal benefits of reducing emissions are given by d . Using b) marginal cost are given by $\frac{1}{3}E_1 = \frac{1}{2}E_2$. As $E_2 = R - E_1$, we get $\frac{1}{3}E_1 = \frac{1}{2}(R - E_1) \rightarrow E_1 = \frac{3}{5}R$. Equating marginal benefit to marginal cost, we get $d = \frac{1}{3}E_1 = \frac{1}{5}R$, so that $R = 5d$. Hence, the government sets a Cap of $2000 - 5d$ ton CO2. The price that arises for a permit equals d (if $R = 5d$, then $E_1 = 3d$, so that marginal cost in equilibrium equal d).

This corresponds perfectly to theory: marginal external cost of emissions are constant, and equal to d . Hence, a Pigouvian tax (or emission fee) would be equal to d . Under Cap-and-trade with the same total emissions as would arise under a (Pigouvian) emission fee, the price for a permit equals the (Pigouvian) emission fee.

Note: You can also find the optimal R by maximising social efficiency $dR - \frac{1}{6}(E_1)^2 - \frac{1}{4}(E_2)^2$ with respect to E_1 and E_2 , subject to $R = E_1 + E_2$.

e) Now, the simplest approach is to maximise social efficiency $dR - \frac{1}{6}(E_1)^2 - \frac{1}{4}(E_2)^2$ with respect to R , subject to $E_1 = \frac{1}{2}R$ and $E_2 = \frac{1}{2}R$. After substituting for E_1 and E_2 , we differentiate $dR - \frac{1}{6}(\frac{1}{2}R)^2 - \frac{1}{4}(\frac{1}{2}R)^2$ to R , and get first-order condition

$$d - \frac{1}{12}R - \frac{1}{8}R = 0 \rightarrow R = \frac{24}{5}d$$

Hence, the optimal total reduction is smaller than under Cap-and-trade. As Cap-and-trade yields a more efficient way of reducing emissions than demanding equal reductions, it is optimal to reduce emissions further under Cap-and-trade.

Exercise 2.5

a) Note that the demand function gives the private marginal benefit of education: For each level of q , we have $MPB = P$. Socially efficient q : $P + MEB = MC$, which gives $100 - 2q + 25 - \frac{1}{2}q = 20 \rightarrow q^* = 42$.

Privately optimal q (no intervention): $P = MC$ gives $100 - 2q = 20 \rightarrow q^P = 40$.

Note: this exercise is not about some firm maximising profit. Hence, maximising profit of some imaginary monopolist by maximising $\pi = Pq - cq = (100 - 2q)q - 25q$ with respect to q is a completely wrong approach to the exercise.

b) Pigouvian subsidy is equal to marginal damage benefit at social optimum, so equal to MEB at q^* . At $q = 42$, $MEB = 4$ = Pigouvian subsidy. As $q^* = 42$, total expenditures on the subsidy are $42 \cdot 4 = 168$.

c) Total efficiency (= total value created by the market) is the sum of consumer surplus, producer surplus, (total!) external effects, and subsidy cost.

Two ways of calculating: using integration or using graph. Make sure you understand what you do and why!

Using integration: Without subsidy, $q^P = 40$. With subsidy: price = 0, so $q^S = 50$. Net marginal benefit of education as a function of q is given by $MPB + MEB - MC = 100 - 2q + 25 - \frac{1}{2}q - 20 = 105 - 2\frac{1}{2}q$. Hence, effect on efficiency is given by $\int_{40}^{50} (105 - 2\frac{1}{2}q) dq = 105(50 - 40) - \frac{5}{4}((50)^2 - (40)^2) = -75$.

Using a graph: Without subsidy, $q^P = 40$. Price equal to constant marginal cost implies no producer surplus. Consumer surplus equals $\frac{1}{2}(100 - 20)40 = 1600$. Total external benefits from education is the area below the MEB-curve. This equals $\frac{1}{2}(25 - 5)40 + 5 \cdot 40 = 600$. (note: area below MEB-curve can be split into a triangle and a square; you can also integrate the MEB-function). Hence, total efficiency without subsidy = 2200.

With subsidy: price = 0, so $q^S = 50$. Consumer surplus equals $\frac{1}{2}(100 - 0)50 = 2500$. Total external benefits are $\frac{1}{2}(25 - 0)50 = 625$. Subsidy cost are $20 \cdot 50 = 1000$. Hence, total efficiency is $2500 + 625 - 1000 = 2125$.

Conclusion: Free university education yields an efficiency loss of 75 (or: implies that the value created by the market reduces by 75).

Explanation: without any subsidy, consumption is too low from social perspective. With free education, there is overconsumption ($q^S = 50 > 42 = q^*$). Given the numbers in this exercise, the latter inefficiency is larger than the former.

Beyond the question: maximal efficiency arises in the social optimum. Pigouvian subsidy of 4 yields $q = 42$, consumer surplus then equals $\frac{1}{2}(100 - 16)42 = 1764$. Total external benefits = $\frac{1}{2}(25 - 4)42 + 4 \cdot 42 = 609$. Subsidy cost equal $4 \cdot 42 = 168$. Hence

total welfare equals $1764 + 609 - 168 = 2205$.

d) When $P = a - 2q$, we get $q^p = \frac{a-20}{2}$ and $q^* = \frac{a+5}{2\frac{1}{2}}$. Pigouvian subsidy equals MEB at q^* : $25 - \frac{a+5}{5}$. Hence, the Pigouvian subsidy decreases in a . The reason is that here, the marginal external benefits of education decrease in q . Hence, when higher private benefits imply that more people take more education, this reduces the marginal positive externality in equilibrium, yielding an optimally smaller Pigouvian subsidy.

Beyond the question: note that this implies that if the returns to education increase, this does not immediately imply that education should be subsidized more!

Exercise 2.6

a) In the Nash equilibrium, citizens' travel time by car is equal to travel time by bike: $40 = 25 + \frac{1}{100}(C - 500) \rightarrow C = 2000$.

b) The socially efficient number of cars can be found by minimizing total travel time: $\min_C (5000 - C)40 + C(25 + \frac{1}{100}(C - 500))$, which yields first-order condition $\frac{1}{50}C - 20 = 0 \rightarrow C^* = 1000$. This is due to the Tragedy of the Commons / a negative externality: each individual who chooses car does not take into account that this implies that other car users spend more time in the traffic jam.

c) Proposal car lanes: New Nash eq: $40 = 25 + \frac{1}{100}(C - 2000) \rightarrow C = 3500$. Travel time: still 40 minutes for everyone, so no reduction in travel time. Proposal bike lanes: New Nash eq: $35 = 25 + \frac{1}{100}(C - 500) \rightarrow C = 1500$. Everyone now has travel time of 35 minutes. Total reduction in travel time: $5000 * (40-35) = 25000$ minutes. The key point here is that citizen adjust their behavior! If choices would remain the same as under a), the extra car lanes would fully remove traffic jams such that the 2000 car drivers would spend only 25 minutes. Yet, that makes driving by car faster than driving by bike, such that a subset of citizens does change behavior and decides to take the car. Ultimately, this eliminates any gains from this policy. This in contrast to the improved bike lanes, which also benefit car drivers (in equilibrium) as it reduces the number of car drivers.

Note: Even more car lanes (such that there is no traffic jam with 5000 cars) would be a possible way to overcome the Tragedy of the Commons, but is of course costly.

d) Both types of intervention yields the same outcome: citizens' choice between bike and car is socially efficient. This holds by definition of Pigouvian subsidies and taxes. Another way of arguing is that as this is a binary choice, the relative cost and benefit of each choice matters, such that it does not matter whether choice A becomes more attractive or choice B less attractive.

Exercise 2.7

a) False. In equilibrium, 6 inhabitants will buy a sheep. Saving yields after one year 1100 euro. Hence, an inhabitants will buy a sheep if and only if this yields at least

1100 euro. Hence, in the Nash equilibrium, the price of a sheep must be equal to 1100 euro. This happens when there are 6 sheep.

b) True. Total value of the investment of all 10 inhabitants is $P(S)*S + (10-S)*1100$, where $P(S)$ is the price of a sheep as given and $10 - S$ is the number of inhabitants that saves. Substituting for $P(S)$ and maximising to S gives $S^* = 3$.

c) False. Tragedy of the commons arises when resources are rival and non-excluded.

d) True. With a fee of F per sheep, there will be S sheep, where S follows from $1700 - F - 100S = 1100 \rightarrow S = 6 - \frac{F}{100}$ (the choice of independent inhabitants

between saving and a sheep). Total income of the poor inhabitant is $F * (6 - \frac{F}{100})$.

Maximising gives first-order condition $6 - \frac{F}{50} = 0 \rightarrow F = 300$. This yields 3 sheep, and total income is 900 euro. Alternative reasoning using Coase Theorem: Property rights are now assigned, so the outcome will be socially efficient: 3 sheep (question b). Maximal fee such that there will be 3 sheep is 300 euro.

e) True. No-one of the rich inhabitants is worse off (all still have 1100 euro after one year, as under question a) and the poor inhabitant is better off. Hence, social welfare increases, given the additive SW-function.

f) True. In case of a Pareto-improvement (as in case of questions d and e) no one gets lower utility while at least one person gets higher utility. With an additive SW function, this results in higher social welfare.